Calciner Kiln BK-1854

Client: Sample Client **Plant / Site:** Specialty Chemicals Plant
Equipment identified: 1854 Calciner, 24" OD x 31'-0" Long, ABB Raymond / Combustion Engineering / FLSmidth
OEM documents: 1854 Calciner Drawings.pdf, B-508898-1 CA.pdf, B-508898-1 REV. B_CA.pdf, L-8543-1272 Shell GA drawing.pdf
CMMS export: 1854-bom-export.csv (47 rows)
In CMMS for this asset: 47 entries under "BK-1854 CALCINER" — coverage figures below measure how much of the OEM documents these entries cover, not whether the BOM is empty.

32%	47	22	47	12	1
DRAWING COVERAGE	IN CMMS FOR ASSET	MATCHED	MISSING FROM CMMS	CRITICAL MISSING	NEEDS REVIEW

Document coverage: The drawings cover the mechanical calciner assembly (shell, drive, trunnions, seals); the furnace heater and control electronics in the CMMS are documented separately.

12 critical parts are missing from the CMMS — immediate attention recommended.

7 components identified by AI visual inspection of engineering drawings — listed in no parts table.

Missing from CMMS — Critical (12)

`FA-110-22-997` **Fixed Seal Ring (feed end, assembly LA-110-22-996)** CRITICAL

Risk: The fixed seal ring is a primary sealing component at the feed end; its absence from CMMS means no stocked replacement for a critical wear part.

Recommendation: Add FA-110-22-997 to CMMS with qty 1 minimum stock and link to the fixed seal assembly BOM.

Sourcing: OEM proprietary · typical lead time 12-26 weeks — OEM-engineered seal component; ABB Raymond calciner parts now route through FLSmidth — verify the part number is still active before an outage forces the issue.

`FA-110-22-998` **Fixed Seal Ring Wear Plate (feed end and discharge end seal assemblies)** CRITICAL

Risk: Wear plates are consumable contact surfaces in the seal assembly; without stocked replacements, seal failure leads to process gas leakage and unplanned downtime.

Recommendation: Add FA-110-22-998 to CMMS with qty 2 minimum stock (one per seal end) and assign to seal assembly PM task.

Sourcing: Fabricate from drawing · typical lead time 3-6 weeks — Flat wear plate a machine shop can produce from the seal assembly drawing if OEM supply is slow or obsolete.

`MP-29317` **Trakrol 3" Hi-Temp roller (seal assembly, 6 per end)** CRITICAL

Risk: Hi-temp rollers support the rotating cylinder against the fixed seal; wear or failure causes seal misalignment and process gas leakage.

Recommendation: Add MP-29317 to CMMS with qty 12 minimum stock (6 per seal end) and include in seal inspection PM.

FC-77-4-393 **Riding Ring 42" OD x 2.5" Face (Qty 2, AISI 1045)** CRITICAL

Risk: Riding rings are primary wear components supporting the rotating cylinder on trunnion rolls; failure causes catastrophic cylinder collapse.

Recommendation: Add FC-77-4-393 to CMMS as a long-lead critical spare with qty 1 minimum and procurement lead time noted.

FA-110-6-163 **Trunnion Roll Assembly with Bearings (Qty 2, feed end side)** CRITICAL

Risk: Trunnion roll assemblies support the full weight of the rotating cylinder; bearing failure causes cylinder misalignment and potential catastrophic damage.

Recommendation: Add FA-110-6-163 to CMMS as a critical spare with qty 1 minimum and link to trunnion PM inspection interval.

FA-110-6-164 **Trunnion Roll Assembly with Bearings (Qty 2, discharge end side)** CRITICAL

Risk: Trunnion roll assemblies support the full weight of the rotating cylinder; bearing failure causes cylinder misalignment and potential catastrophic damage.

Recommendation: Add FA-110-6-164 to CMMS as a critical spare with qty 1 minimum and link to trunnion PM inspection interval.

LA-82-477 **Thrust Roll Assembly 6"x2" Face (Qty 2)** CRITICAL

Risk: Thrust rolls control axial movement of the cylinder; failure allows the cylinder to walk off the trunnions causing catastrophic damage.

Recommendation: Add LA-82-477 to CMMS as a critical spare with qty 1 minimum and include in trunnion PM.

FA-110-22-919 **Fixed Seal Ring (cylinder assembly, Qty 2)** CRITICAL

Risk: Fixed seal rings at both ends prevent process gas leakage; absence from CMMS means no stocked replacement for a critical wear item.

Recommendation: Add FA-110-22-919 to CMMS with qty 2 minimum stock and link to seal inspection PM.

K1854-01 **Brass Seal Ring, 35"OD x 28"ID x 1" thick, with grease grooves (K1854 calciner)** CRITICAL

Risk: The brass seal ring is a precision-machined wear component at the cylinder ends; without a spare, seal failure causes extended unplanned downtime.

Recommendation: Add K1854-01 to CMMS as a critical long-lead spare with qty 1 minimum and note 6-8 week lead time for custom machining.

Emergency Drive Unit (Wisconsin Natural Gas Engine) with associated drive components CRITICAL

Risk: The emergency drive prevents cylinder damage during power outages by keeping the cylinder rotating; without spare parts the emergency drive may fail when needed.

Recommendation: Identify Wisconsin engine model, create CMMS equipment record, and stock critical engine spare parts (spark plugs, belts, filters).

Found by drawing inspection: Visible in View A-A of drawing L-8543-1272 and L-8643-2755 as 'Emergency Drive Unit, Wisconsin Natl Gas Engine' connected to the cylinder drive train; critical safety device with no parts in CMMS.

Motion Sensor (cylinder rotation detection)

CRITICAL

Risk: The motion sensor detects cylinder rotation and triggers the emergency drive on loss of rotation; failure defeats the emergency drive interlock.

Recommendation: Verify CMMS entry 9195583 (SICK distance sensor) covers this function, or add the specific motion sensor model to CMMS.

Found by drawing inspection: Labeled 'MOTION SENSOR' in the lower left of the plan view on drawing L-8543-1272 and L-8643-2755; appears to be a proximity or tachometer-type sensor mounted near the cylinder drive end.

Thermocouple (T/C) assemblies (discharge end and furnace zones)

CRITICAL

Risk: Thermocouples provide temperature feedback for furnace control; failure causes loss of process temperature control and potential product quality issues or equipment damage.

Recommendation: Add thermocouple assemblies to CMMS with qty 2 minimum stock and link to furnace temperature control PM.

Found by drawing inspection: Labeled 'T/C' at multiple locations in View F-F and the plan view of drawing L-8543-1272; thermocouples are visible at the discharge end and furnace zones as critical process instruments.

Missing from CMMS — Important (18)

`HA-110-22-299` Clamp Ring (fixed seal assembly, both ends)

IMPORTANT

Risk: The clamp ring retains the graphite seal ring in position; loss or damage without a spare causes extended seal downtime.

Recommendation: Add HA-110-22-299 to CMMS with qty 2 minimum stock.

Sourcing: Fabricate from drawing · typical lead time 2-4 weeks — Simple machined ring; fabricable from the drawing by a local machine shop.

`GP-2688` Antiseize Thread Lube (cylinder assembly fasteners)

IMPORTANT

Risk: OEM specifies antiseize on all fasteners per drawing note; absence risks galling of high-temperature fasteners during maintenance.

Recommendation: Add GP-2688 or equivalent antiseize compound to CMMS and include in calciner PM task list.

`SA-77-4-392` Side Plate, riding ring mounting hardware (Qty 24)

IMPORTANT

Risk: Side plates are part of the riding ring mounting hardware assembly; missing spares delay riding ring replacement.

Recommendation: Add SA-77-4-392 to CMMS with qty 24 minimum stock as part of riding ring hardware kit.

`SJ-77-4-372` Top Plate, riding ring mounting hardware (Qty 12)

IMPORTANT

Risk: Top plates are part of the riding ring mounting hardware assembly; missing spares delay riding ring replacement.

Recommendation: Add SJ-77-4-372 to CMMS with qty 12 minimum stock as part of riding ring hardware kit.

`SE-77-4-373` Middle Plate, riding ring mounting hardware (Qty 12)

IMPORTANT

Risk: Middle plates are part of the riding ring mounting hardware assembly; missing spares delay riding ring replacement.

Recommendation: Add SE-77-4-373 to CMMS with qty 12 minimum stock as part of riding ring hardware kit.

SH-77-4-374 **Side Bar, riding ring mounting hardware (Qty 24)** IMPORTANT

Risk: Side bars are part of the riding ring mounting hardware assembly; missing spares delay riding ring replacement.

Recommendation: Add SH-77-4-374 to CMMS with qty 24 minimum stock as part of riding ring hardware kit.

SA-77-18-188 **Knocker (Qty 4, cylinder assembly)** IMPORTANT

Risk: Knockers prevent material buildup inside the cylinder; worn or broken knockers reduce calciner efficiency and can cause plugging.

Recommendation: Add SA-77-18-188 to CMMS with qty 4 minimum stock and include in annual PM inspection.

SA-77-18-189 **Knocker Pin (Qty 4, cylinder assembly)** IMPORTANT

Risk: Knocker pins are wear items that retain the knocker assembly; failure causes knocker loss inside the cylinder.

Recommendation: Add SA-77-18-189 to CMMS with qty 4 minimum stock and pair with knocker PM task.

Tire Filler Plates, carbon steel, 5"x2.5", 18-11/16" radius, in thicknesses 1/4", 3/16", 1/8", 1/16" (Qty 8 per thickness) IMPORTANT

Risk: Filler plates maintain the riding ring-to-shell fit as the tire wears; without correct shim stock the riding ring will rock and accelerate trunnion wear.

Recommendation: Add tire filler plates in all four thicknesses to CMMS with qty 8 per thickness and link to riding ring inspection PM.

Tire Keeper, 3/4" sq bar, 4" wide, 18-11/16" radius, carbon steel (Qty 16) IMPORTANT

Risk: Tire keepers prevent the riding ring from migrating axially off the shell; loss causes riding ring to walk and contact the trunnion housing.

Recommendation: Add tire keepers to CMMS with qty 16 minimum stock and include in riding ring PM inspection.

Item 8 (B-508898-1) **Sprocket Mounting Flange (Qty 1, shell assembly)** IMPORTANT

Risk: The sprocket mounting flange is the structural interface between the drive chain and the cylinder shell; damage without a spare causes extended downtime.

Recommendation: Add sprocket mounting flange to CMMS as a long-lead spare with OEM drawing reference B-508898-1.

Item 5 (B-508898-1) **Spiral Flights, 3/16" x 2" x 12", Qty 3 (feed end spiral flights)** IMPORTANT

Risk: Spiral flights move material through the feed end of the cylinder; worn or missing flights reduce throughput and cause material backup.

Recommendation: Add spiral flights to CMMS with qty 3 minimum stock and include in annual cylinder inspection.

Item 6 (B-508898-1) **Agitating Flights, PL 3/16" x 2" x 12", Qty 120 (96 heated, 24 cooling sections)** IMPORTANT

Risk: Agitating flights are wear items that tumble material for heat transfer; worn flights reduce calcination efficiency and product quality.

Recommendation: Add agitating flights to CMMS with qty 24 minimum stock (one full row) and include in annual cylinder inspection.

`Item 2 (B-508898-1)` **Dam Ring, PL 3/4" x 24 7/8"OD x 8"LG, 316 SS** IMPORTANT

Risk: The dam ring controls material bed depth in the cylinder; damage or loss affects residence time and product quality.

Recommendation: Add dam ring to CMMS as a fabricated spare with drawing reference B-508898-1 and note 316 SS material requirement.

`Item 4 (B-508898-1)` **Feed End Dam Ring, PL 3/16" x 23 5/8"OD x 17 1/2"LG, 310 SS** IMPORTANT

Risk: The feed end dam ring retains material at the feed end; damage causes material spillage and reduced calciner efficiency.

Recommendation: Add feed end dam ring to CMMS as a fabricated spare with drawing reference B-508898-1 and note 310 SS material requirement.

`Item 10 (B-508898-1)` **End Plate, PL 1/4" x 24"OD x 28"OD, 316L SS (Qty 2)** IMPORTANT

Risk: End plates seal the cylinder ends and interface with the seal assemblies; corrosion or damage without a spare causes extended downtime.

Recommendation: Add end plates to CMMS as fabricated spares with drawing reference B-508898-1 and note 316L SS material requirement.

Guard with Graphite Block (trunnion roll guard assembly) IMPORTANT

Risk: The graphite block guard lubricates the riding ring contact surface; without replacement graphite blocks the riding ring and trunnion rolls experience accelerated wear.

Recommendation: Add graphite block (guard type) to CMMS and verify CMMS entries 6427781/6427782 cover this specific application.

Found by drawing inspection: Labeled 'GUARD W/ GRAPHITE BLOCK TYP.' in Half Section B-B of drawing L-8543-1272 and L-8643-2755; graphite block is a consumable lubrication element visible at each trunnion roll position.

Adjusting Screws (trunnion base, discharge end) IMPORTANT

Risk: Adjusting screws set the trunnion roll position relative to the riding ring; loss or damage prevents proper cylinder alignment adjustment.

Recommendation: Add adjusting screws to CMMS with qty 4 minimum stock and include in trunnion alignment PM procedure.

Found by drawing inspection: Labeled 'ADJUSTING SCREWS' in the plan view near the discharge end trunnion base on drawing L-8543-1272; visible as threaded adjustment mechanisms on the trunnion support frame.

Missing from CMMS — Routine (17)

`MP-12991` **Buthd Screw .31-16 x 1.00 brass (seal assembly, 6 per end)** ROUTINE

Risk: Brass screws secure the graphite ring and are consumed during each seal replacement; absence delays seal changeout.

Recommendation: Add MP-12991 to CMMS with qty 12 minimum stock and include in seal replacement kit.

`13-3041` **HXHD Screw .38-16 x 1.25 (seal assembly, 8 per end)** ROUTINE

Risk: Fasteners for the seal assembly clamp ring; not stocked means delays during seal maintenance.

Recommendation: Add 13-3041 to CMMS with qty 16 minimum stock or include in a seal hardware kit.

13-3180 **HXHD Screw 1.00-8 x 8.00 (seal assembly, 6 per end)** ROUTINE

Risk: Large fasteners securing the fixed seal ring assembly; non-standard size unlikely to be available locally.

Recommendation: Add 13-3180 to CMMS with qty 12 minimum stock.

15-1009 **Plain Washer 1.00 (seal assembly, 12 per end)** ROUTINE

Risk: Washers for the 1" seal assembly bolts; consumed during each seal service.

Recommendation: Add 15-1009 to CMMS with qty 24 minimum stock or include in seal hardware kit.

11-2002 **Hex Nut 1.00-8UNC (seal assembly, 6 per end)** ROUTINE

Risk: Nuts for the main seal assembly bolts; consumed during each seal service.

Recommendation: Add 11-2002 to CMMS with qty 12 minimum stock or include in seal hardware kit.

11-4002 **Jam Nut 1.00-8UNC (seal assembly, 6 per end)** ROUTINE

Risk: Jam nuts lock the main seal assembly bolts; consumed during each seal service.

Recommendation: Add 11-4002 to CMMS with qty 12 minimum stock or include in seal hardware kit.

MP-12161 **Locknut 1.25-12UNF (seal assembly, 6 per end)** ROUTINE

Risk: Locknuts for the Trakrol roller studs in the seal assembly; non-standard fine thread unlikely to be locally available.

Recommendation: Add MP-12161 to CMMS with qty 12 minimum stock.

13-3116 **Scr Cap HH .63-11x3.75 (cylinder assembly, Qty 7)** ROUTINE

Risk: Cap screws for the cylinder assembly; non-standard length may not be locally available causing maintenance delays.

Recommendation: Add 13-3116 to CMMS with qty 7 minimum stock.

13-3131 **Scr Cap HH .75-10x3.25 (cylinder assembly, Qty 10)** ROUTINE

Risk: Cap screws for the cylinder assembly; non-standard length may not be locally available causing maintenance delays.

Recommendation: Add 13-3131 to CMMS with qty 10 minimum stock.

15-2008 **Lock Washer .63 (cylinder assembly, Qty 4)** ROUTINE

Risk: Lock washers for cylinder assembly fasteners; consumed during maintenance.

Recommendation: Add 15-2008 to CMMS with qty 4 minimum stock or include in cylinder hardware kit.

15-2007 **Lock Washer .75 (cylinder assembly, Qty 10)** ROUTINE

Risk: Lock washers for cylinder assembly fasteners; consumed during maintenance.

Recommendation: Add 15-2007 to CMMS with qty 10 minimum stock or include in cylinder hardware kit.

11-2008 **Hex Nut .63 (cylinder assembly, Qty 4)** ROUTINE

Risk: Hex nuts for cylinder assembly fasteners; consumed during maintenance.

Recommendation: Add 11-2008 to CMMS with qty 4 minimum stock or include in cylinder hardware kit.

11-2000 **Hex Nut .75 (cylinder assembly, Qty 10)** ROUTINE

Risk: Hex nuts for cylinder assembly fasteners; consumed during maintenance.

Recommendation: Add 11-2000 to CMMS with qty 10 minimum stock or include in cylinder hardware kit.

733 **Cotter Pin 1/8 x 1.25 (knocker pin retention, Qty 4)** ROUTINE

Risk: Cotter pins retain the knocker pins; loss causes knocker assembly to fall into the cylinder.

Recommendation: Add cotter pin 1/8 x 1.25 to CMMS with qty 8 minimum stock.

Item 12 (B-508898-1) **Hex HD Bolts 3/4"-UNC x 3 1/2"LG, SAE GR.8, with Hex Nuts and Flat Washers (riding ring hardware)** ROUTINE

Risk: Grade 8 riding ring mounting bolts are safety-critical fasteners; non-standard length and grade may not be locally available.

Recommendation: Add riding ring bolt kit (bolts, nuts, washers) to CMMS with qty 12 minimum set and link to riding ring PM.

Sight Hole with Plug (furnace section, 1 per zone) ROUTINE

Risk: Sight hole plugs seal the furnace inspection ports; missing or damaged plugs cause heat loss and potential burn hazard.

Recommendation: Add sight hole plug (pipe plug, appropriate material for furnace temperature) to CMMS with qty 2 minimum stock.

Found by drawing inspection: Labeled 'SIGHT HOLE W/PLUG 1 PER ZONE' in Section C-C of drawing L-8543-1272; visible as threaded plug assemblies on the furnace shell, 2 zones shown.

Pipe Nipples and Pipe Plugs (furnace shell purge connections, 2 required) ROUTINE

Risk: Purge connection fittings maintain furnace atmosphere integrity; missing spares delay maintenance of the purge system.

Recommendation: Add 3/4" pipe nipples and plugs (high-temp rated) to CMMS with qty 4 minimum stock.

Found by drawing inspection: Labeled '2 NIPPLES 2 REQ'D' in Section C-C of drawing L-8543-1272 on the furnace shell; visible as small threaded fittings used for atmosphere purge connections.

Lead times are planning estimates based on sourcing category — not supplier quotes.

Needs Review — drawing and CMMS disagree (1)

K1854-01 **Brass Seal Ring, 35"OD x 28"ID, 1" thick, for K1854 calciner**

CMMS has: 9174292 — DNR-SEAL,RING,1854,36"OD,27.5"ID,1"THK

Issue: Both are 1854 calciner seal rings, but the OEM drawing says 35"OD x 28"ID while the CMMS entry says 36"OD x 27.5"ID — a wrong-size ring ordered during a seal outage will not fit.

Recommendation: Measure the installed ring or check the drawing revision, then correct either the CMMS dimensions or stock the drawing's size.

Matched Parts (22)

OEM PART	CMMS ENTRY	CONFIDENCE
9174072 — Trunnion, 10" OD x 4" wide, through-hardened	9174072 — RP-TRUNNION,10"ODx4"WIDE,THRU-HARD-ENED	100%
9174073 — Trunnion, 10" DIA x 8" wide, 2" shaft	9174073 — RP-TRUNNION,10"DIA.x 8"WIDE,2"SHAFT	100%
MP-11256 — Compression Spring M-70 (seal assembly)	9239675 — SPRING,COMPRESSION,M-70,MP-11256	100%
DI-7860 — 120# Roller Chain, 180 ft	9107576 — K-CHAIN,ROLLER,#120 COTTERED	85%
DI-7877 — Offset Link/Coupler, #120 chain	9107598 — LINK,OFFSET,#120	90%
DI-7877-A — Connecting Link, #120 chain	9107613 — LINK,CONNECTING,#120	90%
VP-41290-K — Drive Sprocket (cylinder drive)	9232644 — NS-SPROCKET,120A84,31"ID,41"OD,CS1040	75%
MP-11502 — Idler Sprocket	9232641 — NS-SPROCK-ET,120B11H,1-1/2"BORE,W/NDLEBRG	70%
FA-77-3-210 / Item 11 (B-508898-1) — Sprocket 84T, split sprocket bracket (1045 steel)	7510045 — NS-SPROCKET,120B21H,2-1/4" BORE	55%
HA-110-22-817 / HA-110-22-918 — Bellows Seal Assembly FE and DE	9234618 — NS-FE,BELLOW,FA-110-21-177	70%
HA-110-12-198 — Graphite Seal Ring (fixed seal assembly)	9173201 — SEAL,GRAPHITE,35"OD,28"ID,1-5/8"THK	85%
9233552 — Thrust Roll Assembly (single assembly)	9233552 — NS-ROLLER,THRUST,SINGLE ASSY,1854	100%
FA-110-22-997 — Fixed Seal Ring (feed end and discharge end)	9233190 — NS-KIT,SEAL,FIREBOX,37"ODX25"ID,GEM-COLIT	55%
GP-2688 — Antiseize Thread Lube	9104 — K-SEALANT,SILICONE,RTV,RED,12/CASE	0%
Y-2926 — Permatex #2 Sealant (seal assembly)	9108206 — K-SEALANT,SILICONE,RTV,RED,12/CASE	55%
MP-29317 — Trakrol 3" Hi-Temp (seal assembly roller)	9195583 — P-SENSOR,DIS-TAN,SICK,DL100-23AA2110,4PIN	0%
9179700 — Bearing, pillow block, 2", Dodge 066302, non-expansion	9179700 — P-BEARING,P.B.,2",DODGE,066302,NON-EXP	100%
9179701 — Bearing, pillow block, 2", Dodge 066308, expansion	9179701 — P-BEARING,P.B.,2",DODGE,066308,EXP	100%
9107746 — Furnace Heater Element, ABB FC-11031-276	9107746 — HEATER,FURNACE,ABB,FC-11031-276	100%
9109388 — Motor, 5HP, 230/460V, 1800RPM, 184TC Frame (cylinder drive)	9109388 — K-MOTOR,5,230/460V,1800RPM,184TC FRAME**	100%
9170530 — Gear Reducer, Sumitomo (drive assembly)	9170530 — GEAR,REDUCER,SUMITOMO	100%
9170804 — Coupling Spider, Lovejoy L100 (drive assembly)	9170804 — COUPLING SPIDER, LOVEJOY/099-L100	100%

In CMMS but not in OEM documents (7)

9189720 **AIR CONDITIONER, PANEL - HOFFMAN MCLEAN**

Panel air conditioner for the electrical control cabinet; not referenced in any OEM mechanical drawing but likely required for the SCR/furnace control panel.

9164170 **R-CONTROLLER, SCR,WATLOW DT3481204A15AAA**

SCR temperature controller for furnace heating elements; not listed in OEM mechanical drawings but is an electrical/controls component for the furnace assembly.

9163613 **NS-RECTIFIER,SCR, 480V, 75AMP**

SCR power rectifier for furnace heating circuit; electrical component not referenced in OEM mechanical drawings but required for furnace operation.

6415161 **BASE,15x14x3 BLACK, CORTINA**

Traffic safety base (Cortina brand); appears to be a safety/housekeeping item unrelated to calciner mechanical components.

6415162 **CONE CHANNELIZER 49" CORTINA**

Traffic safety cone (Cortina brand); appears to be a safety/housekeeping item unrelated to calciner mechanical components.

6428075 **LUBE,CONCENTRATE LIQ,GRAPHITE,IKD WBL-52**

Liquid graphite lubricant concentrate; likely used for riding ring/trunnion lubrication but not specifically called out in OEM drawings reviewed.

6428074 **NS-TOOL,SPRAY APPLICATOR,1.5 GAL,IKD**

Spray applicator tool for graphite lubricant; a maintenance tool rather than a spare part, included in CMMS likely for procurement tracking.